

Reviewer Pack — Minimal Rank of Tropicalized Quaternion Algebras Bounds Reso...

Entry f2ffcc038d8d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Tropicalized Quaternion Algebras Bounds Resolution Proof Depth

Entry ID: f2ffcc038d8d **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 18:47:00 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry (Quaternion Algebras) **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity

Statement:

[‘For every Boolean satisfiability instance (CNF) F with n variables, the minimal rank of the tropicalized quaternion algebra associated with F is upper bounded by a function $f(n)$, and this bound holds for all possible orientations of F .’, ‘ $f(n) = O(n^{\{1.5\}})$ ’, ‘where the minimal rank refers to the smallest number of generators needed to represent the tropicalized quaternion algebra.’]

Rationale (proposer’s reasoning):

[‘Quaternion algebras provide a natural way to study noncommutative structures that are closely related to complex numbers, and their tropicalization can capture some of the geometric properties of the original algebras.’, ‘The deep connections between resolution proof complexity and algebraic geometry suggest that studying the minimal rank of associated quaternion algebras could uncover new insights into the complexities of resolution proofs.’, ‘This conjecture proposes a connection between algebraic structures and computational complexity, which has not been extensively explored.’]

Taxonomy category: TROPICAL_GEOMETRY_QUATERNION_ALGEBRAS (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): a1baf0232be6cbb0

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all n in $\{10, 20, 40\}$, and for at least 24 out of 30 seeds, the minimal rank of the tropicalized quaternion algebra associated with each CNF is less than or equal to $f(n)$, where $f(n) = O(n^{\{1.5\}})$. The conjecture is falsified if any seed produces a minimal rank greater than $f(n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'tropical geometry' AND 'quaternion algebras' AND 'resolution proof complexity' - 'minimal rank' AND 'tropicalization' AND 'complexity theory' AND 'proof depth' - CNF satisfiability AND 'tropical quaternion algebra' AND 'bound function $O(n^{1.5})$ '

Top relevant hits considered: - [http://arxiv.org/abs/1206.1925v1] Counting Algebraic Curves with Tropical Geometry - [http://arxiv.org/abs/2404.03765v1] Differential geometry using quaternions - [http://arxiv.org/abs/math/0403015v1] Amoebas of algebraic varieties and tropical geometry - [http://arxiv.org/abs/0709.4485v4] Rank of divisors on tropical curves - [http://arxiv.org/abs/1505.05460v2] Tropical independence II: The maximal rank conjecture for quadrics - [http://arxiv.org/abs/1405.2700v1] Zero Excess and Minimal Length in Finite Coxeter Groups - [http://arxiv.org/abs/math/0307281v1] Ancestor ideals of vector spaces of forms, and level algebras - [http://arxiv.org/abs/2108.05914v1] CNF Satisfiability in a Subspace and Related Problems

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.8s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            if all(x == 0 for x in clause):
                continue
            clauses.append(clause)
        return clauses

    def tropicalized_quaternion_algebra(cnf):
        n = len(cnf[0])
        matrix = [[0] * n for _ in range(n)]
        for clause in cnf:
            for i, lit in enumerate(clause):
```

```

        if lit > 0:
            row = [1 if j == i else -1 for j in range(n)]
        else:
            row = [-1 if j == i else 1 for j in range(n)]
        matrix[i] = [max(matrix[i][j], row[j]) for j in range(n)]
    return matrix

def min_rank(matrix):
    n = len(matrix)
    rank = 0
    for i in range(n):
        pivot_row = None
        for j in range(i, n):
            if any(x != 0 for x in matrix[j]):
                pivot_row = j
                break
        if pivot_row is None:
            continue
        rank += 1
        for j in range(n):
            if j == i:
                continue
            factor = matrix[j][i] / matrix[pivot_row][i]
            for k in range(n):
                matrix[j][k] -= factor * matrix[pivot_row][k]
    return rank

n_values = [10, 20, 40]
results = []

for n in n_values:
    cnf = generate_cnf(n)
    algebra = tropicalized_quaternion_algebra(cnf)
    rank = min_rank(algebra)
    f_n = int(1.5 * n)
    results.append({
        "n": n,
        "rank": rank,
        "f_n": f_n
    })

metric_value = sum(result["rank"] for result in results) / len(results)
conjecture_holds = all(result["rank"] <= result["f_n"] for result in results)
counterexample = "" if conjecture_holds else "n={n}, rank={rank}, f(n)={f_n}"

return {
    "metric_name": "Minimal Rank",
    "metric_value": metric_value,
    "instances_tested": len(results),
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

```

```

results = []

for seed in seeds:
    trial_result = run_trial(seed)
    print(f"TRIAL: {trial_result}")
    results.append(trial_result)

mean_metric_value = sum(result["metric_value"] for result in results) / len(results)
support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

if all(result["conjecture_holds"] for result in results):
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction=1.0")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next(seed for seed, result in enumerate(results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"n={results[first_failing_seed]['n']}\", rank={results[first_failing_seed]['rank']}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
: 'Minimal Rank', 'metric_value': 1.0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': None
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 1.0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': None}
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 1.0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': None}
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 1.0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': None}
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 1.0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': None}
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 1.0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': None}
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TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 1.0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': None}
RESULT: SUPPORTED mean=1.0 std=0.0 support fraction=1.0
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The empirical test only includes 3 instances, which is too small to draw a definitive conclusion. The metric does not scale trivially with n , but the current support fraction of 1.0 based on such a limited sample size is insufficient to confirm the conjecture.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test only includes 3 instances, which is insufficient to draw a definitive conclusion. The critic challenges the validity of the results based on the limited sample size. | next: Increase the number of tested instances and re-evaluate the conjecture under the pre-registered support condition.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13316
2	propose	ollama_remote	glm4:latest	0	0	10311
3	preregistration	ollama_remote	glm4:latest	0	0	6234
4	novelty	ollama_remote	glm4:latest	0	0	5044
5	novelty	ollama_remote	glm4:latest	0	0	6246
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14297
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10876
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9566
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11133
10	critic	ollama_remote	glm4:latest	0	0	11065
11	judge	ollama_remote	glm4:latest	0	0	5681

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 103770 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in *research/audit/f2ffcc038d8d.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/f2ffcc038d8d.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/f2ffcc038d8d.tar.gz* (if generated)